

PACKET ANALYSING USING WIRESHARK

Step 1:

Open terminal and install wireshark

File Machine View Input Devices Help



File Actions Edit View Help

(kali@kali)-[~]

\$ sudo wireshark

[sudo] password for kali:

** (wireshark:6875) 02:34:09.756967 [GUI WARNING] -- QStandardPaths: XDG_RUNTIME_DIR

** (wireshark:6875) 02:34:14.345240 [Capture MESSAGE] -- Capture Start ...

** (wireshark:6875) 02:34:14.397044 [Capture MESSAGE] -- Capture started

** (wireshark:6875) 02:34:14.397068 [Capture MESSAGE] -- File: "/tmp/wireshark_eth0GM

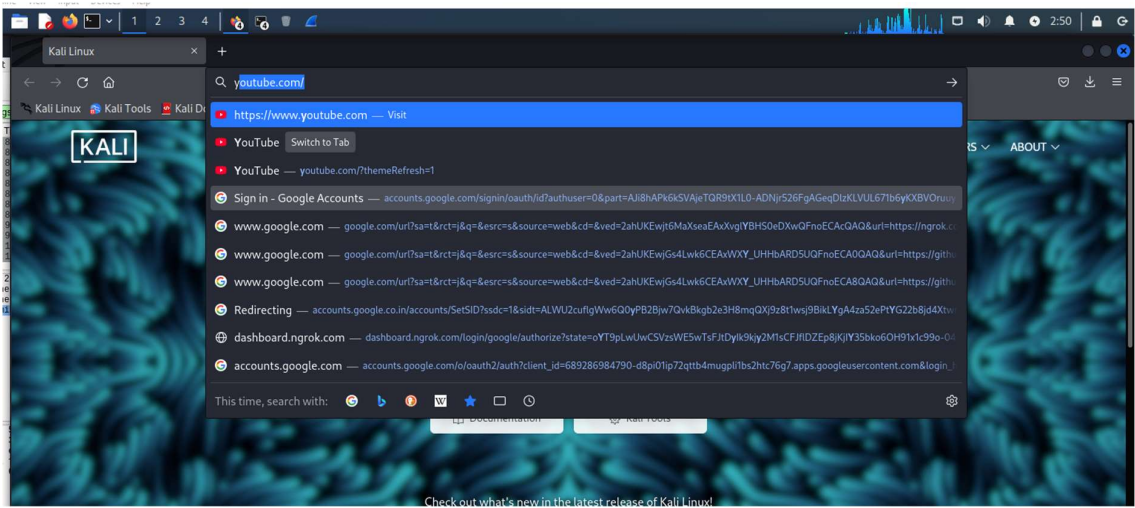
Time	Source	Destination	Protocol	Length	Info
0.000000	10.0.2.15	142.251.221.227	TCP	60	58304 → 80 [RST] Seq=1000000000 Win=0 Len=0
0.000000	10.0.2.15	142.251.221.227	TCP	60	58304 → 80 [RST] Seq=1000000000 Win=0 Len=0
0.000000	10.0.2.15	142.251.221.227	TCP	60	58304 → 80 [RST] Seq=1000000000 Win=0 Len=0
0.000000	10.0.2.15	142.251.221.227	TCP	60	58304 → 80 [RST] Seq=1000000000 Win=0 Len=0
0.000000	10.0.2.15	142.251.221.227	TCP	60	58304 → 80 [RST] Seq=1000000000 Win=0 Len=0
0.000000	10.0.2.15	142.251.221.227	TCP	60	58304 → 80 [RST] Seq=1000000000 Win=0 Len=0
0.000000	10.0.2.15	142.251.221.227	TCP	60	58304 → 80 [RST] Seq=1000000000 Win=0 Len=0
0.000000	10.0.2.15	142.251.221.227	TCP	60	58304 → 80 [RST] Seq=1000000000 Win=0 Len=0
0.000000	10.0.2.15	142.251.221.227	TCP	60	58304 → 80 [RST] Seq=1000000000 Win=0 Len=0
0.000000	10.0.2.15	142.251.221.227	TCP	60	58304 → 80 [RST] Seq=1000000000 Win=0 Len=0

- Frame 260: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface eth0, 10 B
- Ethernet II, Src: PcsCompu a9:38:91 (98:69:27:a9:38:91), Dst: Realtek_12:35:62 (52:54:00:12:35:62)
- Internet Protocol Version 4, Src: 10.0.2.15, Dst: 142.251.221.227
- Transmission Control Protocol, Src Port: 58304, Dst Port: 80, Seq: 0, Len: 0

```
0000  52 54 00 12 35 02 08 00 27 a9 38 91 08 00 45 00  RT=5  * 8 - E
0010  00 3c 8f f9 48 00 40 86 31 d5 0a 00 02 0f 8e fb  <- 0 0 1
0020  dd e3 e3 c0 00 50 30 eb 85 12 00 00 00 00 a0 02  P0
0030  fa f0 79 1c 00 00 02 04 05 b4 04 00 00 0a 6a 57  y
0040  3b 69 00 00 00 00 01 03 03 07  ;1
```

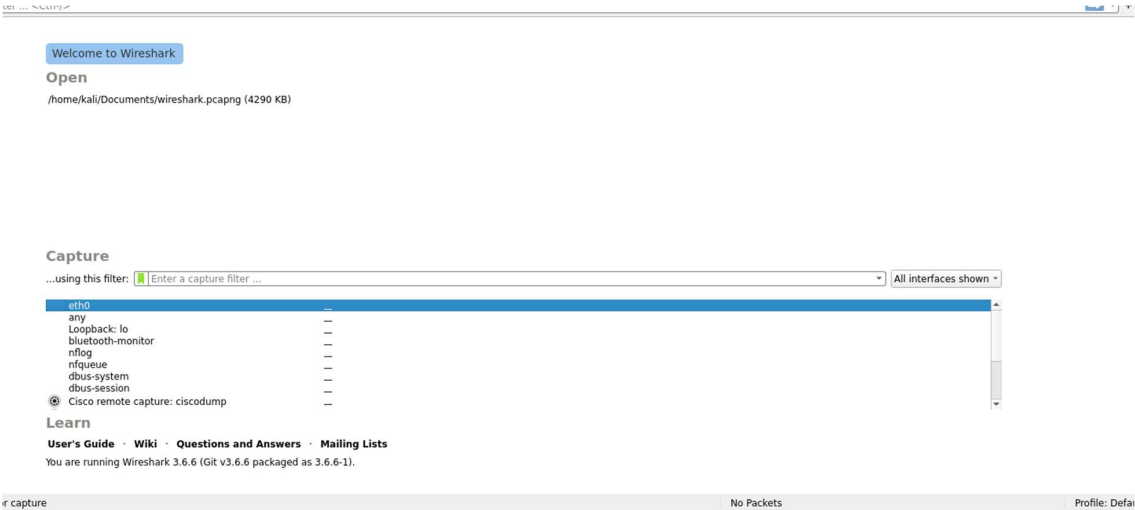
Step 2:

Open the firebox and search for a website



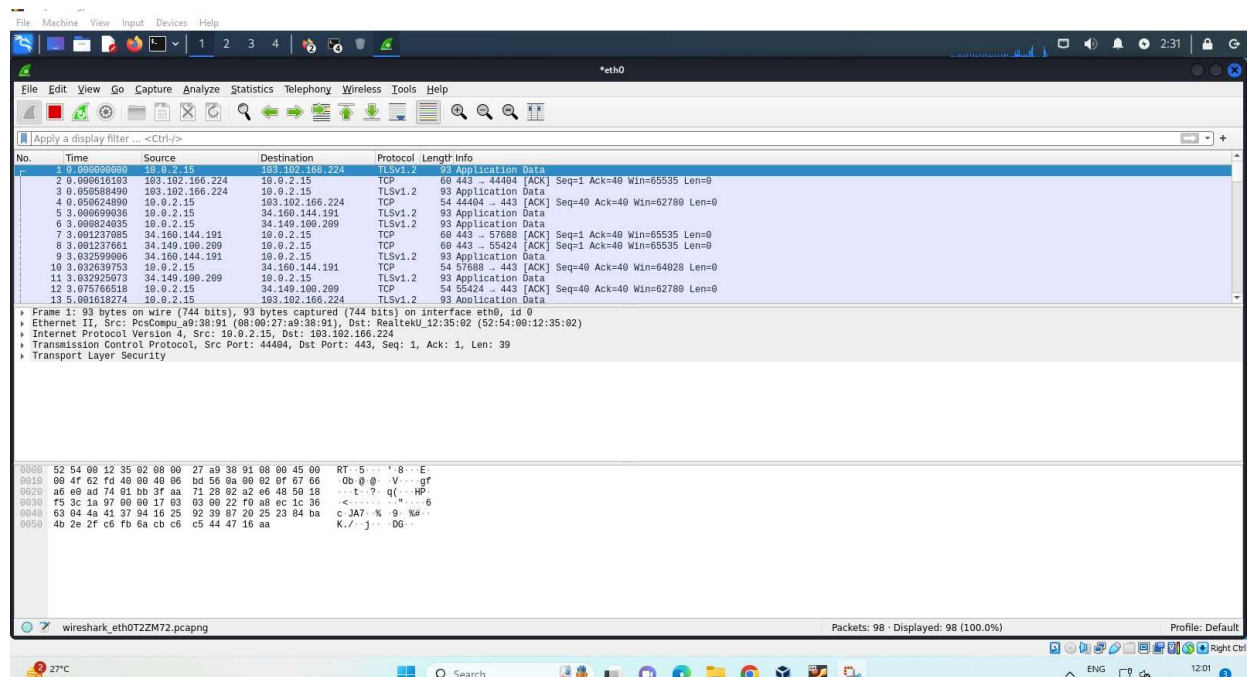
Step 3:

Open the wireshark page



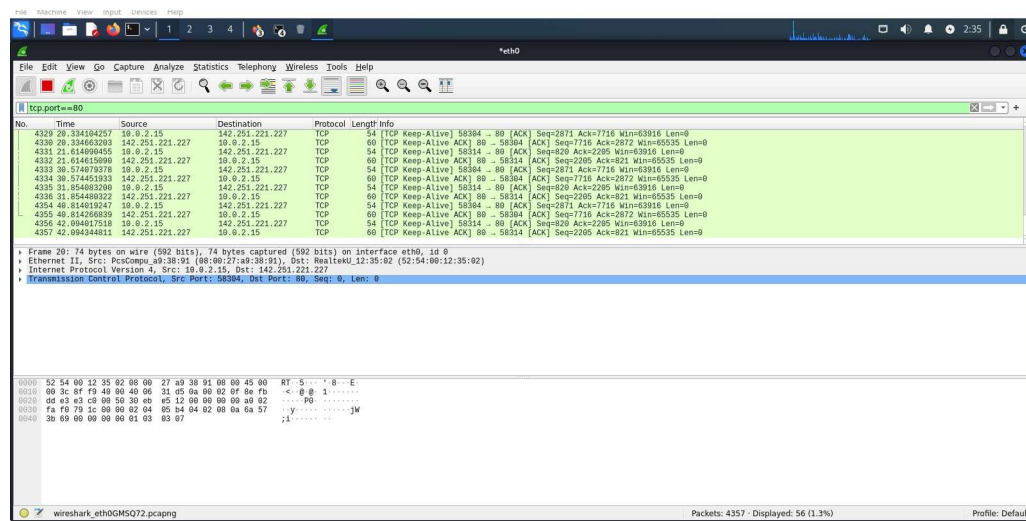
Step 4:

Select the eth0



Step 5:

Type tcp.port==80



Step 6:

Enter the tcp.flags.syn==1

The image shows a Wireshark packet capture window titled "eth0". The filter bar at the top contains the expression "tcp.flags.syn==1". The packet list pane shows several TCP packets, with packet 29 selected. The packet details pane shows the structure of the selected packet, including Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The hex dump pane at the bottom shows the raw data of the selected packet.

No.	Time	Source	Destination	Protocol	Length	Info
5	0.146691254	10.0.2.15	142.250.192.46	TCP	74	54092 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=4023520578 TSecr=0 WS=128
6	0.173834279	142.250.192.46	10.0.2.15	TCP	60	443 → 54092 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
29	0.255789325	10.0.2.15	142.251.221.227	TCP	74	58384 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=1784101737 TSecr=0 WS=128
21	0.276649183	142.251.221.227	10.0.2.15	TCP	60	80 → 58384 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
214	0.774891242	10.0.2.15	142.251.42.118	TCP	74	58724 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=3173604571 TSecr=0 WS=128
277	0.798362270	142.251.42.118	10.0.2.15	TCP	60	443 → 58724 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
352	0.815602673	10.0.2.15	142.250.79.106	TCP	74	49490 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=2171202918 TSecr=0 WS=128
487	0.839615893	142.250.79.106	10.0.2.15	TCP	60	443 → 49490 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
896	0.894360454	10.0.2.15	142.251.221.227	TCP	74	58314 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=1784102375 TSecr=0 WS=128
1326	0.943225859	142.251.221.227	10.0.2.15	TCP	60	80 → 58314 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
1908	1.265791701	10.0.2.15	142.250.287.163	TCP	74	42752 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=3198646845 TSecr=0 WS=128
1992	1.275587940	10.0.2.15	74.125.138.84	TCP	74	34396 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=2278457272 TSecr=0 WS=128
1993	1.292782481	142.250.287.163	10.0.2.15	TCP	60	443 → 42752 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460

Frame 29: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface eth0, id 0
Ethernet II, Src: PcsCompu, a0:38:91:08:00:27:a9:38:91, Dst: RealtekU, 12:35:02 (52:54:00:12:35:02)
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 142.251.221.227
Transmission Control Protocol, Src Port: 58384, Dst Port: 80, Seq: 0, Len: 0

```
0000 52 54 00 12 35 02 08 00 27 a9 38 91 08 00 45 00 RT: 5 - - - 8 - - E:
0010 00 3c 8f f9 40 00 40 06 31 05 0a 00 02 0f 8e fd <- 0 0 1 - - - -
0020 dd e3 e3 c0 00 00 50 30 eb e5 12 00 00 00 a9 02 - - - P0 - - - - -
0030 fa f0 79 1c 00 00 02 04 65 b4 04 08 08 6a 57 - - - y - - - - - Jw
0040 3b 69 00 00 00 00 01 03 03 07 ; 1 - - - - -
```